Course Outline

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I. Introduction

Civilization and The State of the World in the Anthropocene (Year: 2027)

1. The Anthropocene
2. State of the Earth: Year 2027
   (a) The Atmosphere
   (b) The Hydrosphere
   (c) The Cryosphere
   (d) The Biosphere
   (e) The Lithosphere
3. State of the Humans: Year 2027
   (a) Behavioral Organizing Principles
      i. Money
         A. Capital and Wealth
         B. Debt-Based Economics
      ii. Status and Power
      iii. Religion
      iv. Government and Empire
   (b) Technology
   (c) Population
4. State of Civilation
   (a) Globalized
   (b) Weaponized
   (c)

II. Approach

0.1 System Analysis

1. Fundamental Systems Concepts and Perspectives
    (a) System Components and Characterizations
    (b) Systems Science
        i. classical (calculus-based) open/closed systems
        ii. set theory
        iii. graph theory
        iv. network theory
        v. cybernetics
        vi. information theory
        vii. game theory
        viii. complexity theory
        ix. decision theory
        x. queuing theory
        xi. theory of structure (the order of parts) and function (the order of processes)
    (c) Mental Models
2. Spatial Scales
    (a) Global
    (b) Regional
    (c) Local
3. Temporal Scales
    (a) Past
    (b) Present
    (c) Future
4. Analytics
    (a) Probability Models
    (b) Statistical Models
    (c) Agent-Based Models
    (d) Time Series Models

0.2 Opportunity

1. External
   - (a) Creating New Systems
   - (b) Re-vamping Existing Systems
   - (c) The Critical Systems
     - i. Economic
     - ii. Governance
     - iii. Cultural
     - iv. Artistic
     - v. Environmental
     - vi. Existential
       - A. Food
       - B. Water
       - C. Clothing
       - D. Shelter
2. Internal
   - (a) Creativity
   - (b) Intuition
   - (c) Mind, Body, Spirit

III. Topics

0.3 Sustainability Concepts and Frameworks

1. Sustainability Concepts
   - (a) Brundtland Commission
   - (b) UN Sustainable Development Goals
   - (c) Marshall and Toffel
2. Sustainability Frameworks
   - (a) Ecological Sustainability
     - i. Ecosystem Services: Definitions and Metrics
     - ii. Global Ecosystem Health and Ecosystem Service Delivery
   - (b) Social Sustainability
   - (c) Economic Sustainability
   - (d) Agricultural Sustainability

0.4 A Sustainable Biosphere

1. Functional Terrestrial and Aquatic Ecosystems
    (a) Necessary Inputs
        i. Physical Resources
            A. Space
            B. Time
            C. Flow
        ii. Biological Resources
            A. Reliable Primary Productivity
    (b) Necessary Outputs
        i. Measurable Properties
            A. Biodiversity
            B. Energy and Material Recycling
        ii. Enmergent Properties
            A. Adaptive Capacity
            B. Resilience

0.5 A Sustainable Anthroposphere

1. Culture
    (a) Pilars of Belief
        i. Reality and Imagination
        ii. Identity Legacy
    (b) Open Cooperative Competition
        i. Cohesion
        ii. Energy
        iii.
2. Power
    (a) Governance
    (b) Elites
3. Money
    (a) Consumption
        i. Necessary
        ii. Wants
    (b) Value vs Money vs Wealth
    (c) Economy and Trade